

HandsOn12- Serverless Spark ETL Pipeline on AWS

Project Overview

This project is about building an automated data processing system using AWS. The main goal is to process raw CSV data without running anything manually.

In this pipeline, when a file is uploaded, the system automatically processes it using Spark, performs analysis, and stores the results in another location.

Architecture & Data Flow

Data Flow:

S3 (Upload) → Lambda → AWS Glue → S3 (Output)

How it works:

1. First, a CSV file (`reviews.csv`) is uploaded into an S3 bucket.
2. This upload automatically triggers a Lambda function.
3. The Lambda function starts a Glue job.
4. The Glue job reads the file, cleans the data, and runs SQL queries.
5. Finally, the processed data and results are stored in another S3 bucket.

AWS Services Used

Amazon S3

S3 is used to store both input and output data.

- Landing bucket: where raw data is uploaded
- Processed bucket: where results are stored

Create two buckets that are [hhandsonfinallandingg](#) and [hhandsonfinalprocessedd](#)

The screenshot shows the AWS Management Console for the 'us-east-1' region. The left sidebar contains navigation links for Amazon S3, Buckets, Files, Access management and security, Storage management and insights, and Account and organization settings. The main content area is titled 'Buckets' and shows the 'General purpose buckets' tab. A table lists four buckets:

Name	AWS Region	Creation date
aws-glue-assets-848237286339-us-east-1	US East (N. Virginia) us-east-1	April 14, 2026, 12:36:02 (UTC-04:00)
handsonn11cloudcomputing	US East (N. Virginia) us-east-1	April 7, 2026, 11:28:03 (UTC-04:00)
hhandsonfinallandingg	US East (N. Virginia) us-east-1	April 14, 2026, 12:12:10 (UTC-04:00)
hhandsonfinalprocessedd	US East (N. Virginia) us-east-1	April 14, 2026, 12:13:08 (UTC-04:00)

On the right side of the console, there are two panels: 'Account snapshot' (Updated daily) with a 'View dashboard' button, and 'External access summary' (Updated daily) with a description of external access findings.

AWS Glue

Glue is used to process the data using PySpark.

It performs:

- Data cleaning
- Data transformation
- Running SQL queries
- Writing results back to S3

The screenshot displays the AWS Glue Monitoring console. The left sidebar contains navigation links for AWS Glue, Data Catalog, Data Integration and ETL, and Legacy pages. The main content area is titled 'Monitoring' and shows a 'Job runs summary' for the last 7 days. The summary includes: Total runs (4), Running (0), Canceled (0), Successful runs (1), Failed runs (3), Run success rate (25%), and DPU hours (1). Below the summary is a table of job runs (4 total) with columns for Job name, Run status, Type, Start time, End time, Run time, Capacity, Worker type, and DPU hours. The table shows one successful run and three failed runs. At the bottom, there is a 'Resource usage' section with a 'Filter displayed data' dropdown and a 'View service quotas' link.

AWS Glue Monitoring

Start date range: 7 Day

Job runs summary

Total runs	Running	Canceled	Successful runs	Failed runs	Run success rate	DPU hours
4	0	0	1	3	25%	1

Job runs (4)

Job name	Run status	Type	Start time (Local)	End time (Local)	Run time	Capacity	Worker type	DPU hours
process_reviews_job	Succeeded	Glue ETL	04/15/2026 11:40:29	04/15/2026 11:42:22	2 minutes	10	G.1X	0.2
process_reviews_job	Failed	Glue ETL	04/15/2026 10:49:00	04/15/2026 10:50:04	1 minute	10	G.1X	0.1
process_reviews_job	Failed	Glue ETL	04/14/2026 12:38:04	04/14/2026 12:39:00	1 minute	10	G.1X	0.1
process_reviews_job	Failed	Glue ETL	04/14/2026 12:36:14	04/14/2026 12:37:19	1 minute	10	G.1X	0.1

Resource usage

Filter displayed data: Filter data

View service quotas

AWS Lambda & IAM:

The IAM execution role provides the necessary permissions for Lambda to start the Glue job and for Glue to access data from S3. It ensures that all services can interact securely without manual intervention.

The screenshot displays the AWS IAM console interface. The left sidebar shows the navigation menu with 'Identity and Access Management (IAM)' selected. The main content area shows the details for the role 'start_glue_job_triggerrrr-role-87a5leoo'. The 'Summary' section includes the creation date (April 15, 2026, 10:59 UTC-04:00), the last activity (2 hours ago), the ARN (arn:aws:iam::848237286339:role/service-role/start_glue_job_triggerrrr-role-87a5leoo), and the maximum session duration (1 hour). The 'Permissions' tab is active, showing a table of permissions policies. The table has columns for 'Policy name', 'Type', and 'Attached entities'. Two policies are listed: 'Allow-Glue-StartJobRun' (Customer inline, 0 attached entities) and 'AWSLambdaBasicExec...' (Customer managed, 1 attached entity). The 'Permissions boundary' section shows '(not set)'.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/details/start_glue_job_triggerrrr-role-87a5leoo?section=permissions

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
 - Resource analysis
 - Unused access
 - Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

start_glue_job_triggerrrr-role-87a5leoo

Summary

Creation date
April 15, 2026, 10:59 (UTC-04:00)

Last activity
2 hours ago

ARN
arn:aws:iam::848237286339:role/service-role/start_glue_job_triggerrrr-role-87a5leoo

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2)

You can attach up to 10 managed policies.

Filter by Type
All types

☐	Policy name	Type	Attached entities
☐	Allow-Glue-StartJobRun	Customer inline	0
☐	AWSLambdaBasicExec...	Customer man...	1

Permissions boundary (not set)

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Spark SQL Queries Implemented

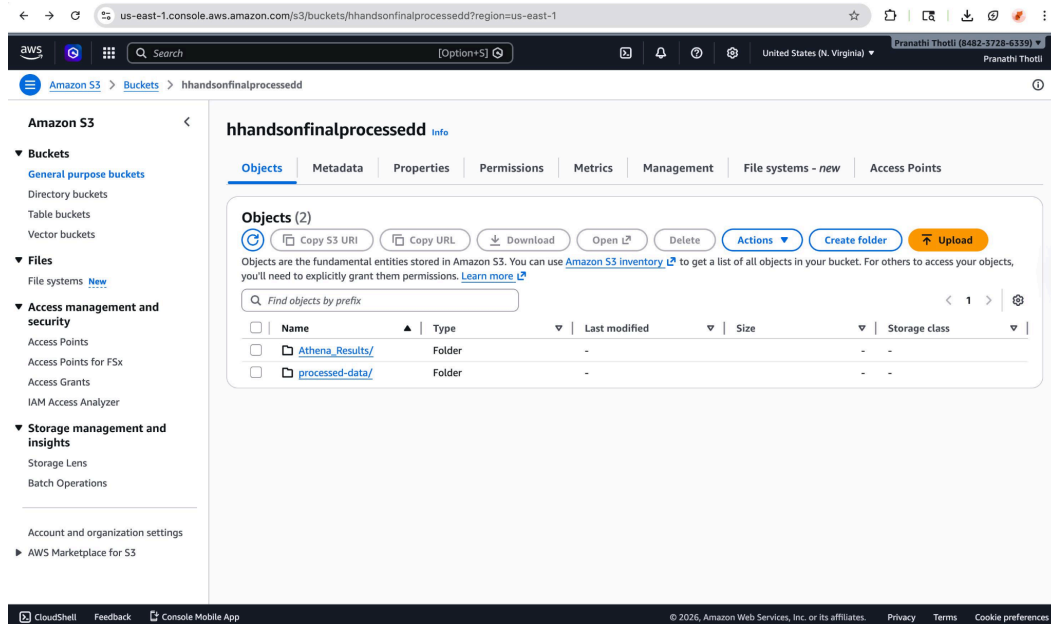
Within the AWS Glue Spark session, several analytical queries were created and executed to analyze the data.

- **Average Rating and Review Count per Product:**
This query calculates the average rating for each product and also counts how many reviews each product has, using the standardized product_id.
- **Date-wise Review Count:**
This query shows the total number of reviews submitted on each day, helping to understand how review activity changes over time.
- **Top 5 Most Active Customers:**
This query finds the top five customers who have given the highest number of reviews and also shows their average rating.
- **Overall Rating Distribution:**
This query displays how many reviews fall under each rating level (1 to 5 stars) and shows their distribution.

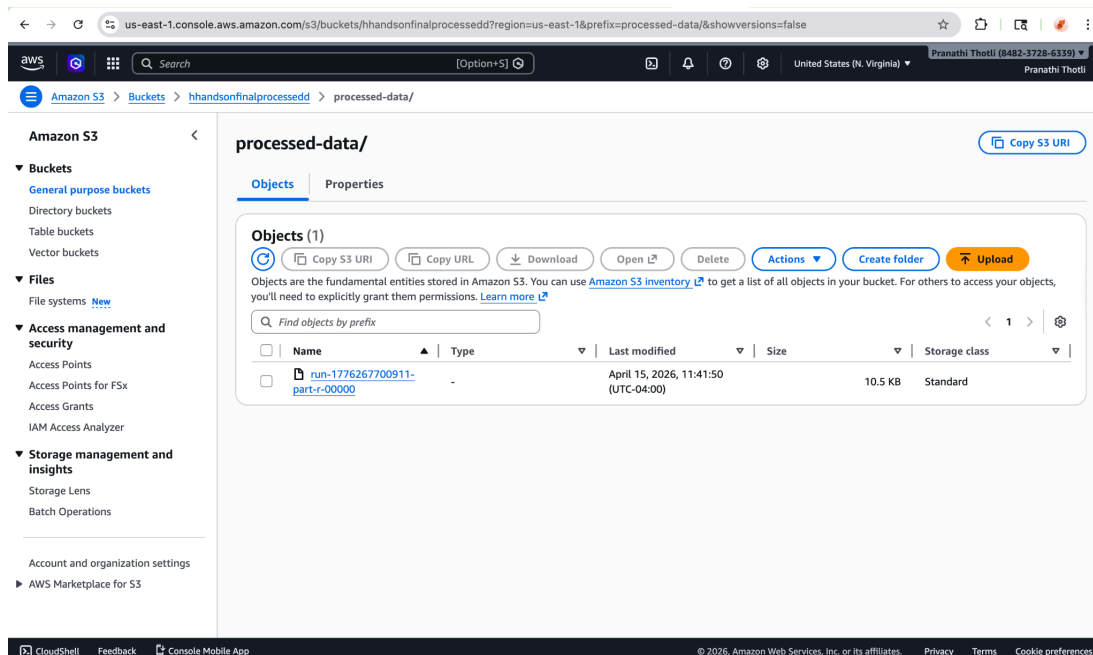
Execution & Results

The pipeline was successfully executed on AWS. The output data was generated correctly and stored in the S3 processed bucket.

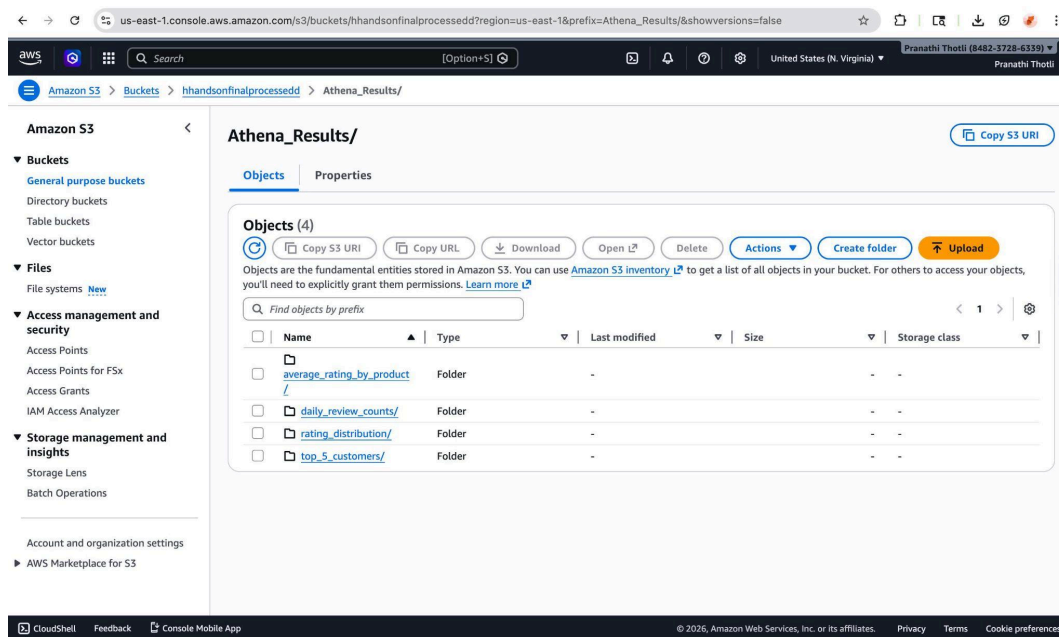
Folders partitioned for Data and Athena Analysis:



Cleaned Transactional Dataset (processed-data):



Analytical Query Outputs (Athena Results):



CleanUp:

The S3 buckets (hhandsonfinallandingg and hhandsonfinalprocesseddd) were first emptied and then deleted.

The Lambda function (start_glue_job_trigger) was removed.

The AWS Glue job (process_reviews_job) was deleted.

The IAM role (AWSGlueServiceRole-Reviews) used for the project was also removed.

